

Bacterial Alkanesulfonate Uptake and FMN-Dependent Desulfonation: A Review-Style Synthesis of the SsuABC–SsuE–SsuD Module

1. Executive Summary

Under sulfur limitation, many bacteria scavenge sulfur from organosulfonates — a large pool of sulfur that is otherwise inaccessible because the sulfur is locked in a chemically inert carbon–sulfur bond. The **alkanesulfonate utilization (ssu) module** is a compact, three-part molecular device that solves this problem. An ABC-type importer, **SsuABC**, delivers aliphatic sulfonates from the periplasm into the cytoplasm; a flavodoxin-like reductase, **SsuE**, uses NADPH to generate reduced flavin mononucleotide (FMNH₂); and a luciferase-family TIM-barrel monooxygenase, **SsuD**, consumes that FMNH₂ together with molecular oxygen to cleave the C–S bond of the sulfonate, releasing **sulfite** (which feeds cysteine biosynthesis) and the corresponding **aldehyde**. The module is transcriptionally gated: in *Escherichia coli* the *ssuEADCB* operon is induced only during sulfate/cysteine starvation, under the control of the LysR-type regulators **CysB** (master) and **Cbl** (accessory).

Mechanistically, this is a **two-component flavin-dependent monooxygenase** system in which the flavin is not a tightly bound prosthetic group but a **diffusible substrate/cosubstrate** that is produced by one enzyme (SsuE) and consumed by another (SsuD). SsuD strongly prefers reduced flavin (FMNH₂ K_d ≈ 0.32 μM vs. oxidized FMN K_d ≈ 10.2 μM) and enforces an **obligately ordered** reaction: reduced flavin binds first, then the alkanesulfonate, and only then does O₂ react at the flavin C4a position to form a **C4a-(hydro)peroxyflavin**, the oxygenating intermediate that attacks the sulfonate carbon. A dynamic active-site loop shields the reduced flavin from unproductive oxidation, and the flavin is delivered from SsuE to SsuD via a **transient, oligomerically regulated protein–protein complex** rather than being released freely into bulk solvent — although whether transfer is by free diffusion or true channeling remains an open question.

The two enzymes are structurally and evolutionarily **unrelated ancient scaffolds** (a flavodoxin-like reductase and a luciferase-like TIM barrel) that have been convergently paired into a functional module. The same physiological end — release of sulfite for cysteine biosynthesis — is achieved in parallel by a **chemically distinct** route: the taurine dioxygenase **TauD**, an α -ketoglutarate/Fe(II)-dependent enzyme with its own transporter (TauABC). The module therefore sits at the intersection of flavin biochemistry, ABC transport, and sulfur-starvation regulation, and its boundaries are best drawn to include transport, flavin supply, and desulfonation, while excluding downstream sulfite assimilation and the parallel taurine/dioxygenase system.

2. Definition and Biological Boundaries

2.1 What is included

The system reviewed here is the **reusable bacterial sulfur-scavenging module** comprising three functional steps:

1. **Alkanesulfonate import** — the SsuABC ABC transporter (periplasmic binding protein SsuA, ATPase SsuB, permease SsuC) moves aliphatic sulfonates across the inner membrane into the cytoplasm.
2. **Reduced FMN supply** — the FMN reductase SsuE reduces FMN to FMNH₂ at the expense of NADPH.
3. **Oxidative desulfonation** — the monooxygenase SsuD uses FMNH₂ + O₂ to release sulfite from the imported alkanesulfonate, producing an aldehyde.

Two obligatory internal dependencies define the module's wiring: **SsuABC** → **SsuD** (the transporter supplies cytoplasmic substrate to the monooxygenase) and **SsuE** → **SsuD** (the reductase supplies reduced flavin to the monooxygenase cycle). Both converge on SsuD, which is the catalytic heart of the system.

2.2 What should be treated separately

Several neighboring processes are frequently discussed alongside this module but fall **outside** its boundary and are mechanistically distinct:

- **Sulfite assimilation.** The sulfite released by SsuD is subsequently reduced and incorporated into cysteine by the sulfate/cysteine biosynthetic machinery. That downstream assimilation is a separate pathway and is not part of the desulfonation module.
- **The taurine/TauD route.** Under the same starvation cue, *E. coli* co-induces a parallel operon whose desulfonation enzyme, **TauD**, is an α -ketoglutarate/Fe(II)-dependent **dioxygenase** — completely different chemistry from the flavin monooxygenase SsuD (see Finding F005). TauD preferentially acts on **taurine** (2-aminoethanesulfonate), which SsuD does **not** accept. TauD, its transporter TauABC, and taurine metabolism should be treated as a sibling system, not as part of the ssu module.
- **Substrate-specific transport paralogs and lineage variants.** The *Pseudomonas aeruginosa* **msu** operon (MsuE/MsuD) is a homologous but distinct module tuned to methanesulfonate and using NADH rather than NADPH. It illuminates the ssu module by comparison but is a separate paralogous system.
- **Sulfur-limitation regulation.** CysB/Cbl-mediated transcriptional control is the switch that turns the module on; it is a regulatory input to the module rather than one of its mechanistic steps, and per the working scope it is treated as adjacent context.

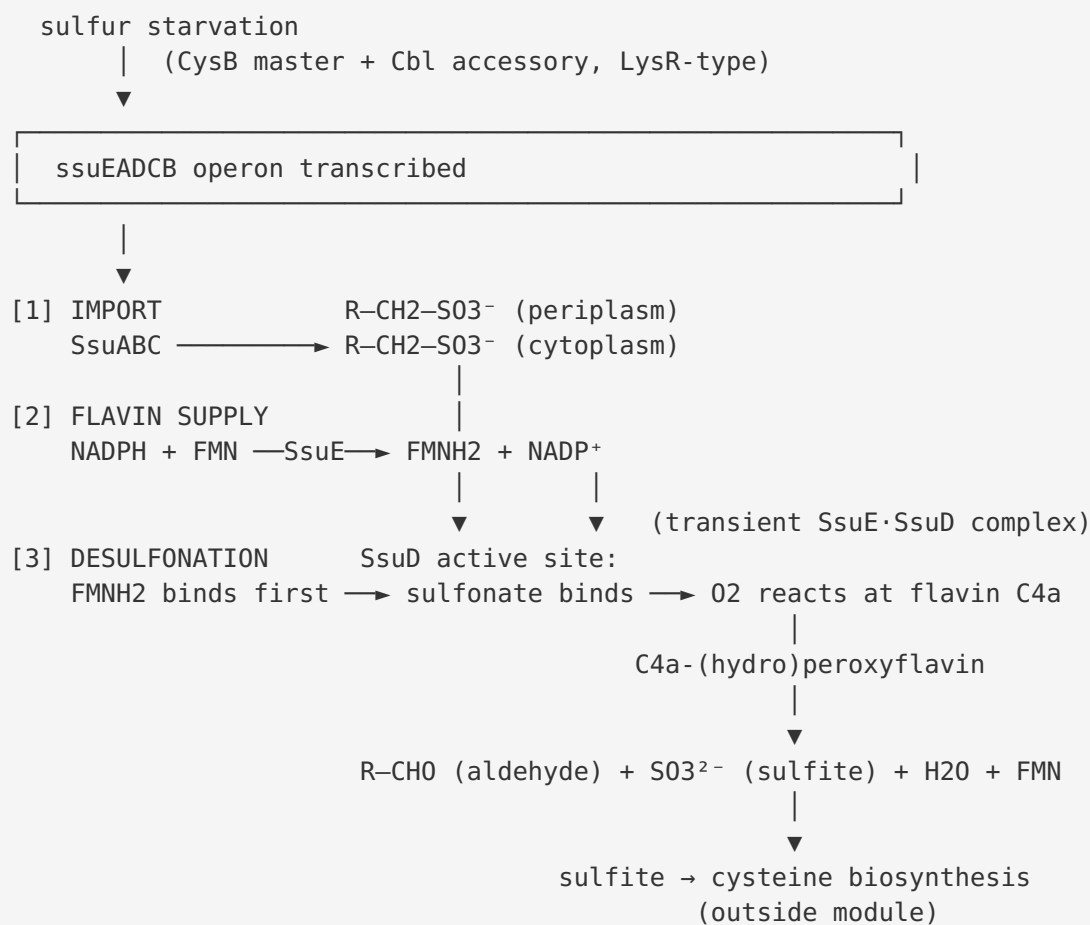
2.3 Competing definitions

The principal definitional tension in the literature is whether "the alkanesulfonate monooxygenase system" refers narrowly to the **two-enzyme SsuE/SsuD flavin-transfer pair** or more broadly to the **entire ssuEADCB operon including transport**. Biochemical/structural papers tend to use the narrow sense (the SsuE–SsuD couple and its flavin chemistry), whereas physiological/genetic papers use the broad sense (transport + desulfonation + regulation). This review adopts the broad, module-level definition — transport, flavin supply, and desulfonation — because the three steps are co-regulated and functionally interdependent, while explicitly excluding downstream assimilation and the parallel TauD chemistry.

3. Mechanistic Overview

3.1 The best current model, step by step

The module operates as an ordered relay in which sulfur, reducing equivalents, and oxygen are brought together at the SsuD active site:



3.2 Obligatory, conditional, and accessory steps

- **Obligatory:** (i) reduction of FMN by SsuE; (ii) binding of reduced flavin to SsuD *before* the sulfonate; (iii) formation of the C4a-(hydro)peroxyflavin as the oxygenating species; (iv) O₂ as cosubstrate. The **substrate binding order is obligatory** — reduced flavin must precede the alkanesulfonate — and this order is catalytically essential rather than merely kinetically preferred (Finding F002).

- **Conditional / regulated:** the **kinetic mode of flavin handoff**. In isolation, SsuE follows an ordered sequential mechanism; in the presence of SsuD plus octanesulfonate, it shifts to a rapid-equilibrium-ordered mechanism with a ~10-fold increase in FMN K_m (Finding F004). The oligomeric state of SsuE (tetramer $\rightleftharpoons$ dimer) is modulated by FMN/NADPH and governs transfer. These features are conditional on partner and substrate presence.
- **Accessory / protective:** a **dynamic active-site loop** in SsuD that shields the reduced flavin from unproductive autoxidation. It is not part of bond cleavage per se, but its deletion abolishes activity because the reduced flavin is lost to non-productive oxidation (Finding F002).

3.3 Molecular assemblies carrying out each step

Step	Player	Fold / family	Catalytic role
Import	SsuABC	ABC-type transporter (binding protein + ATPase + permease)	ATP-driven uptake of aliphatic sulfonates
Flavin supply	SsuE	Flavodoxin-like (FMN reductase/quinone reductase/WrbA) superfamily; dimer-of-dimers tetramer	NADPH-dependent reduction of FMN $\rightarrow$ FMNH ₂
Desulfonation	SsuD	Luciferase-family TIM barrel; homotetramer	FMNH ₂ /O ₂ -dependent oxygenolytic C–S cleavage

4. Major Molecular Players and Active Assemblies

4.1 Finding F001 — A co-transcribed, starvation-induced three-part operon

The defining organizational feature of the module is that a **single operon encodes transporter, reductase, and monooxygenase together**, and that this operon is switched on by sulfur starvation. In *E. coli*, starvation for sulfate/cysteine induces the *ssuEADCB* genes, encoding the ABC transporter (SsuABC), the FMN reductase (SsuE), and the monooxygenase (SsuD). This induction requires the LysR-type transcriptional regulators **CysB** (the master regulator of the cysteine regulon) and **Cbl** (an accessory regulator). The parallel *tau* cluster is

similarly organized and similarly CysB-dependent. The homologous **msuEDC** operon of *P. aeruginosa* is likewise CysB-dependent but is tuned to methanesulfonate, demonstrating that the same operon logic is reused across lineages with different substrate preferences.

Evidence: "Starvation for sulfate leads to the expression of the tauABCD and ssuEADCB genes. Each of these gene clusters encodes an ABC-type transport system required for uptake of aliphatic sulfonates and a desulfonation enzyme." — PMID: 11479697. Regulatory dependence: "Expression of the tau and ssu genes requires the LysR-type transcriptional regulatory proteins CysB and Cbl." The lineage-specific *msu* paralog: "The MsuD protein catalyzed the desulfonation of alkanesulfonates, requiring oxygen and FMNH₂ for the reaction, and showed highest activity with methanesulfonate. MsuE was an NADH-dependent flavin mononucleotide (FMN) reductase, which provided reduced FMN for the MsuD enzyme." — PMID: 10049377.

4.2 Finding F002 — Ordered catalysis through a C4a-(hydro)peroxyflavin

SsuD is a flavin-dependent monooxygenase that uses reduced flavin as a **cosubstrate**, not a prosthetic group. It binds FMNH₂ with roughly 30-fold higher affinity than oxidized FMN (**K_d $\approx$ 0.32 $\pm$ 0.15 μ M for FMNH₂ vs. 10.2 $\pm$ 0.4 μ M for FMN, 1:1 stoichiometry**), which is the thermodynamic basis for an **ordered mechanism** in which reduced flavin must bind before the alkanesulfonate. Once reduced flavin and sulfonate are both bound, molecular oxygen reacts at the flavin C4a position to generate a **C4a-(hydro)peroxyflavin**, the oxygenating intermediate that attacks the sulfonate carbon to cleave the C–S bond. This intermediate can be observed spectroscopically (absorbance near 370 nm) and accumulates specifically when FMNH₂ is delivered to SsuD in the presence of substrate rather than being premixed and oxidized away; its formation correlates with product (octanal) formation. A **dynamic loop** guards the reduced flavin against unproductive oxidation; deletion variants of this loop fail to protect the reduced flavin and lose activity, explaining mechanistically why ordered binding and controlled flavin delivery matter.

Evidence: "The SsuD enzyme shows a clear preference for FMNH₂ (K_d, 0.32 $\pm$ 0.15 μ M) compared to FMN (K_d, 10.2 $\pm$ 0.4 μ M) with a 1:1 binding stoichiometry for each form of the flavin." — PMID: 18198899. "A C4a-(hydro)peroxyflavin is the

oxygenating intermediate in the mechanism of desulfonation by the alkanesulfonate monooxygenase." — [PMID: 21880344](#). Loop protection: "the SsuD deletion variants failed to protect reduced flavin from unproductive oxidation" — [PMID: 22564769](#).

4.3 Finding F003 — Two unrelated folds: a luciferase-like TIM barrel and a flavodoxin-like reductase

The two catalytic enzymes come from **entirely different structural lineages**. SsuD crystallizes as a **homotetramer**, each subunit a **TIM-barrel** fold enlarged by four insertion regions that mediate intersubunit contacts. Despite low sequence identity, SsuD is structurally related to **bacterial luciferase** and to an **archaeal coenzyme F₄₂₀-dependent reductase**, placing it firmly in the luciferase-like flavin-dependent monooxygenase family. SsuE, by contrast, belongs to the **flavodoxin-like** (NADPH:FMN reductase / quinone reductase / WrbA) superfamily; it assembles as a **dimer-of-dimers tetramer** and carries a distinctive **π -helix** at the tetramer-building interface. This π -helix, generated by insertion of Tyr118 into an otherwise canonical α 4 helix, is a diagnostic feature of reductases from two-component monooxygenase systems and underlies SsuE's unusual **flavin-free** character — mutating Tyr118 (e.g., Y118A) converts the normally flavin-free enzyme into a flavin-bound, canonical flavoprotein-like form.

Evidence: "each subunit being composed of a TIM-barrel fold enlarged by four insertion regions that contribute to intersubunit interactions. SsuD is structurally related to a bacterial luciferase and an archaeal coenzyme F(420)-dependent reductase" — [PMID: 12445781](#). "A π -helix present at the tetramer building interface is unique to the reductases from two-component monooxygenase systems." — [PMID: 24816272](#). "generation of the Y118A SsuE variant converted the typically flavin-free enzyme to a flavin-bound form" — [PMID: 27806563](#).

4.4 Finding F004 — Regulated, transient flavin transfer via a protein–protein complex

Because the flavin is diffusible, the module faces a delivery problem: reduced flavin is chemically fragile and must reach SsuD before it autoxidizes. The solution is a **transient physical complex** between SsuE and SsuD. Affinity chromatography and cross-linking demonstrate a stable SsuE–SsuD complex, and fluorescence measurements suggest very tight association (reported K_d in the low-nanomolar range, proposed 1:1). This partnership

reprograms the reductase: in the presence of SsuD and octanesulfonate, SsuE's kinetic mechanism shifts from ordered sequential to **rapid-equilibrium-ordered**, and its FMN K_m rises ~10-fold — a signature of regulated, on-demand flavin production. Transfer is further controlled by **oligomeric state**: SsuE shifts from tetramer toward dimer in response to FMN or NADPH, whereas the *P. aeruginosa* paralog MsuE shifts in the opposite direction (dimer → tetramer with FMN) and is unresponsive to NADPH — a clear lineage divergence in how flavin transfer is gated. Whether the reduced flavin is passed by **direct channeling** or by **rapid free diffusion** within the complex remains debated across two-component systems.

Evidence: "the results from affinity chromatography and cross-linking experiments support the formation of a stable complex between the flavin mononucleotide (FMN) reductase (SsuE) and monooxygenase (SsuD)" — [PMID: 16997955](#). "in the presence of SsuD and octanesulfonate the kinetic mechanism of SsuE is altered to a rapid equilibrium ordered mechanism, and the K_m value for FMN is increased 10-fold" — [PMID: 15882995](#). "The oligomeric state of SsuE was converted from a tetramer to a dimer/tetramer equilibrium in the presence of FMN or NADPH" — [PMID: 37651343](#).

4.5 Finding F005 — Two chemically distinct routes to the same outcome

The ssu module is best understood alongside its **chemical alternative**. Under identical starvation conditions, *E. coli* co-induces two parallel desulfonation operons that both release sulfite for cysteine biosynthesis but by fundamentally different chemistry. **TauD** is an **α -ketoglutarate/Fe(II)-dependent dioxygenase** that preferentially desulfonates **taurine**, while **SsuD** is an **FMN₂/O₂-dependent monooxygenase** that acts on a broad range of **non-taurine** aliphatic sulfonates. Each is served by its own ABC transporter (TauABC vs. SsuABC) with overlapping but non-identical substrate ranges, and the specificities of transport and desulfonation are matched. This is a textbook example of **convergent function via divergent mechanism**: two evolutionarily unrelated chemistries partitioned by substrate to cover the organism's sulfonate landscape.

Evidence: "The TauD protein is an alpha-ketoglutarate-dependent dioxygenase that preferentially liberates sulfite from taurine (2-aminoethanesulfonic acid). SsuD is a monooxygenase that catalyzes the oxygenolytic desulfonation of a range of aliphatic

sulfonates other than taurine." — [PMID: 11479697](#). "The range of substrates transported by these two uptake systems was largely reflected in the substrate specificities of the TauD and SsuD desulfonation systems." — [PMID: 10781534](#).

5. Evolutionary and Cell-Biological Variation

5.1 Variation across lineages

The module is **modular and portable**, recurring across Gammaproteobacteria (*E. coli*), other Proteobacteria (*P. aeruginosa*), and, based on soil metagenomics, across Actinobacteria and Alphaproteobacteria. Key axes of variation are:

- **Substrate preference of the monooxygenase.** SsuD in *E. coli* accepts a broad range of C2–C10+ alkanesulfonates but not taurine; the MsuD paralog in *P. aeruginosa* is tuned to short-chain **methanesulfonate**.
- **Reductant specificity.** *E. coli* SsuE is **NADPH**-dependent, whereas *P. aeruginosa* MsuE is **NADH**-dependent (Finding F001).
- **Oligomeric gating of flavin transfer.** SsuE (tetramer → dimer with FMN/NADPH) and MsuE (dimer → tetramer with FMN, NADPH-insensitive) shift oligomeric state in opposite directions, indicating that the flavin-transfer control mechanism has diverged even while the overall reaction has been conserved (Finding F004).
- **π -helix insertional residue.** SsuE uses a Tyr insertion to build its interface π -helix; MsuE uses a His insertion. Swapping these residues does not interconvert their kinetic properties, showing the insertional residue alone does not explain the gain of function (see [PMID: 30171650](#)).

5.2 Physiological-state specificity

The module is **not constitutive**. It is a **starvation-state** program: expression is confined to conditions of sulfate/cysteine limitation and depends on CysB/Cbl. In sulfur-replete cells the pathway is off. This is the dominant "cell-state" variation for a bacterial system without differentiated cell types — the relevant states are metabolic (replete vs. sulfur-starved) rather than developmental.

5.3 Alternative routes to the same outcome

As formalized in Finding F005, the alternative route is **chemical, not merely paralogous**: the non-heme-iron dioxygenase TauD reaches the same end (sulfite release) by α -ketoglutarate-dependent chemistry. Terrestrial community surveys indicate that the **ssuDE and tauD systems are the primary and near-ubiquitous** desulfonation pathways in soil bacteria, while other organosulfonate desulfonation pathways are rare or absent — reinforcing that these two convergent chemistries dominate the sulfonate-sulfur scavenging landscape (see [PMID: 41711914](#)).

5.4 Deepest plausible origin and best representatives

The two enzymes have **independent, ancient origins**. SsuD's TIM-barrel fold and its structural kinship to bacterial luciferase and archaeal F₄₂₀-dependent reductases place it within the deeply conserved **luciferase-like monooxygenase (LLM)** family — an ancient flavin-dependent scaffold that predates and extends far beyond sulfonate metabolism. SsuE belongs to the equally ancient **flavodoxin-like** superfamily of flavin/quinone reductases (WrbA-like). The **module** — the specific pairing of these two lineages for desulfonation — is therefore a case of **convergent assembly of pre-existing folds** rather than co-descent of a single ancestral complex. For understanding the ancestral catalytic role, the *E. coli* SsuD/SsuE pair is the **best-characterized representative** (most complete structural, kinetic, and mechanistic data), while bacterial luciferase serves as the reference point for the SsuD family and canonical flavodoxin-like FMN reductases (including the flavin-bound members that lack the π -helix) serve as the reference for placing SsuE's derived, flavin-free character.

6. Constraints, Dependencies, and Failure Modes

6.1 Ordering constraints (what must happen in what order)

1. **Regulation precedes function.** The operon must be de-repressed (CysB/Cbl, sulfur starvation) before any protein is present.
2. **Import precedes desulfonation.** SsuABC must deliver cytoplasmic sulfonate; SsuD acts only on the imported substrate.
3. **Flavin reduction precedes oxygen chemistry.** SsuE must generate FMNH₂ before SsuD can proceed.

4. **Within SsuD, reduced flavin binds before sulfonate before O₂.** This ordering is obligatory (Finding F002); binding oxidized flavin, or admitting O₂ before both flavin and substrate are in place, is unproductive.

6.2 Mutually exclusive / protected events

- **Productive vs. unproductive flavin oxidation.** Free FMNH₂ reacts with O₂ to give H₂O₂ and oxidized FMN — a futile, potentially damaging side reaction. The system suppresses this via (i) tight, ordered binding of FMNH₂ to SsuD, (ii) the protective dynamic loop, and (iii) transient SsuE→SsuD complex-mediated delivery. Loss of loop function converts the reaction from productive oxygenation to unproductive oxidation (Finding F002).
- **Substrate partitioning between SsuD and TauD.** Taurine is handled by TauD, not SsuD; non-aurine alkanesulfonates by SsuD. The transporters reinforce this partition, and **hybrid transporters are non-functional**, indicating the transport components are matched to their cognate systems (Finding F005).

6.3 Evidence that rules out otherwise plausible paths

- The strong preference for FMNH₂ over FMN (~30-fold) and the requirement that reduced flavin bind first **rule out** a mechanism in which SsuD binds and reduces its own flavin in place; the reductase step is outsourced to SsuE.
- Direct detection of a **stable SsuE–SsuD complex** and the SsuD-dependent reprogramming of SsuE kinetics argue against a purely independent, bulk-diffusion model in which the two enzymes never interact — although they do **not** yet distinguish tight channeling from rapid diffusion within a complex.
- The chemical identity of the oxygenating species as a **C4a-(hydro)peroxyflavin** rules out non-flavin oxygenation chemistries for SsuD and firmly places it among canonical flavin monooxygenases.

6.4 Failure modes

Failure	Molecular cause	Consequence
No induction	Loss of CysB/Cbl or sulfur-replete state	Module silent; no scavenging
No substrate delivery	SsuABC defect / hybrid transporter	SsuD starved of substrate
Flavin wasted	Loop deletion; premixing/oxidation of FMNH ₂	Unproductive H ₂ O ₂ formation, no desulfonation
Wrong reductant	NADPH vs NADH mismatch across lineages	Reduced or absent flavin supply
Wrong substrate	Taurine presented to SsuD	Not turned over (TauD's job)

7. Controversies and Open Questions

- 1. Free diffusion vs. channeling of reduced flavin.** The single most debated mechanistic point. A stable SsuE–SsuD complex exists and partner-dependent kinetic reprogramming is documented (Finding F004), yet whether FMNH₂ is handed directly across a protein–protein interface (channeling) or released and rapidly recaptured (diffusion) is unresolved — and the same debate recurs in related two-component systems such as styrene monooxygenase (StyA/StyB), where FADH transfer remains contested (see [PMID: 38306174](#)).
- 2. Identity of SsuD's catalytic residues.** While the fold, the ordered mechanism, and the C4a-(hydro)peroxyflavin intermediate are established, the specific active-site residues that position substrate, stabilize the peroxyflavin, and effect C–S cleavage are not fully pinned down. Molecular-dynamics/QSAR work implicates conformational gating (e.g., Arg297–Glu20/Asp111 salt bridges driving a closed↔open transition; see [PMID: 22571434](#)), but direct experimental assignment of catalytic residues is incomplete.
- 3. Structure and true substrate range of SsuABC.** The transporter is the least characterized component. There is no high-resolution structure of the assembled SsuABC, and its full substrate spectrum — and how it is partitioned against TauABC — is defined largely by genetics and growth phenotypes rather than by direct binding/transport structural data.

4. **What actually generates the π -helix gain-of-function.** The Tyr118 insertion is necessary but **not sufficient** to build SsuE's interface π -helix or to confer two-component behavior; additional structural adaptations are required, and residue swaps with MsuE do not interconvert kinetics ([PMID: 30171650](#)). The full determinants of the flavin-free, flavin-transferring phenotype remain to be mapped.
5. **Cross-organism comparability.** Much of the deep mechanism (kinetics, structures, intermediates) is from *E. coli*, while substrate/reductant divergence comes from *P. aeruginosa* (msu). Care is needed not to overgeneralize *E. coli* SsuE/SsuD details to all lineages, given the demonstrated divergence in reductant specificity and oligomeric gating.

8. Limitations and Knowledge Gaps

- **Enzyme-centric evidence base.** The strongest data concern the SsuE/SsuD flavin chemistry; the transporter SsuABC is comparatively under-studied structurally, so the "import" arm of the module rests more on genetics than on mechanism.
- **In vitro vs. in vivo flavin dynamics.** Binding constants and kinetic mechanisms are measured in vitro; the actual in-cell flavin pool, competition among flavin-dependent enzymes, and physiological flux are not directly measured here.
- **Organismal breadth.** Deep mechanism is dominated by *E. coli*; broad ecological ubiquity is inferred from metagenomic surveys rather than from mechanistic study of diverse taxa.
- **Regulation treated as adjacent.** By scope, CysB/Cbl regulation is described as the gating switch but not dissected mechanistically; the quantitative logic of induction is outside this module's boundary.

9. Proposed Follow-up Experiments/Actions

1. **Resolve flavin transfer mode.** Use pre-steady-state stopped-flow with rapid mixing, viscosity/dilution dependence, and single-molecule or FRET approaches on the SsuE·SsuD complex to distinguish channeling from rapid free diffusion; test whether crowding/dilution changes coupling efficiency.

2. **Determine an SsuABC structure.** Solve cryo-EM structures of SsuABC (apo, substrate-bound, nucleotide-bound) and perform direct transport/binding assays across a panel of alkanesulfonates to define the true substrate range and its partition against TauABC.
 3. **Assign SsuD catalytic residues.** Combine site-directed mutagenesis (candidates from MD/QSAR such as the Arg297–Glu20/Asp111 network) with trapping/spectroscopy of the C4a-(hydro)peroxyflavin to identify residues that stabilize the intermediate and effect C–S cleavage.
 4. **Dissect the π -helix determinants.** Systematically graft SsuE/MsuE interface segments (beyond the single insertional residue) to map the additional adaptations required for the flavin-free, transfer-competent phenotype.
 5. **Cross-lineage comparative kinetics.** Characterize SsuE/SsuD orthologs from Actinobacteria/Alphaproteobacteria (highlighted by soil metagenomics) to test how conserved the ordered mechanism, reductant specificity, and oligomeric gating are.
 6. **In vivo flux and futile-cycle quantification.** Measure H₂O₂ production and desulfonation flux in wild-type vs. loop-deletion SsuD strains under sulfur starvation to quantify the physiological cost of unproductive flavin oxidation.
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10. Key References

PMID	Short title	Role in this review
11479697	<i>Sulfonate-sulfur metabolism and its regulation in E. coli</i>	Operon structure, starvation induction, CysB/Cbl, SsuD vs TauD chemistry (F001, F005)
10049377	<i>A novel FMNH₂-dependent methanesulfonate sulfonate (msu operon, P. aeruginosa)</i>	Lineage paralog; NADH reductase; methanesulfonate preference (F001)
18198899	<i>Catalytic importance of substrate binding order for SsuD</i>	FMNH ₂ vs FMN affinity; ordered mechanism (F002)
21880344	<i>Mechanism for sulfur acquisition by the alkanesulfonate monooxygenase system</i>	C4a-(hydro)peroxyflavin intermediate (F002)
22564769	<i>Deletional studies of a dynamic loop of SsuD</i>	Loop protects reduced flavin (F002)
12445781	<i>Crystal structure of E. coli SsuD</i>	TIM-barrel/luciferase-family fold (F003)
24816272	<i>Crystal structure of E. coli SsuE; catalytic cycle of flavodoxin-like FMN reductases</i>	Flavodoxin-like fold; diagnostic π -helix (F003)
27806563	<i>Transformation of a flavin-free FMN reductase via the π-helix</i>	Tyr118 π -helix controls flavin-free character (F003)
16997955	<i>Detection of protein–protein interactions in the SsuD system</i>	Physical SsuE–SsuD complex (F004)
15882995	<i>Altered mechanism of the FMN reductase with the monooxygenase</i>	Partner/substrate reprogram SsuE kinetics (F004)
37651343	<i>Oligomeric changes regulate flavin transfer in two-component FMN reductases</i>	Oligomeric gating; SsuE vs MsuE divergence (F004)
10781534	<i>Deletion analysis of E. coli taurine and alkanesulfonate transport</i>	Matched transporter/enzyme specificities (F005)
8808933	<i>Sulfate-starvation-regulated genes; taurine utilization gene cluster</i>	tau operon organization; CysB dependence (context)
30171650	<i>An insertional residue does not explain the π-helix gain-of-function</i>	π -helix determinants beyond Tyr118 (open question)

PMID	Short title	Role in this review
29979040	<i>Functional evaluation of the π-helix in SsuE</i>	π -helix role in flavin transfer/oligomeric change (context)
17289450	<i>Mechanism of flavin reduction in the SsuD system</i>	Microscopic steps of SsuE flavin reduction (context)
22571434	<i>Structural/conformational factors in SsuD activity and specificity</i>	Conformational gating; QSAR of substrates (open question)
16511173	<i>Crystallization of <i>E. coli</i> SsuE</i>	Structural groundwork for SsuE (context)
38306174	<i>Fus-SMO chimeric styrene monooxygenase</i>	Comparative two-component flavin-transfer debate (open question)
41711914	<i>Organosulfonate metabolism in chernozem soil communities</i>	Ecological ubiquity of ssuDE/tauD systems (variation)

Prepared as a review-style synthesis of the bacterial alkanesulfonate uptake and FMN-dependent desulfonation module (SsuABC–SsuE–SsuD), based on 5 confirmed findings and 20 reviewed papers. Claims are anchored to the cited primary literature; uncertainties (flavin-transfer mode, SsuD catalytic residues, SsuABC structure) are flagged explicitly and should not be over-generalized across lineages.

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